

COURSE OUTLINES FOR FIRST SEMESTER

SPA 501: Spherical Astronomy and Geodesy (2+1)

Introduction, The great and small circles, spherical angle and spherical triangle, applications to the Earth, longitude and latitude, basics of spherical trigonometry, the celestial sphere, horizontal and equatorial systems of coordinates, observer's meridian and diurnal motion, circumpolar stars, right ascension, the equation of time. Elements of spherical Astronomy, The celestial sphere, Parallax, Aberration and Precession, Refraction, Orbital Elements and Ephemerides, Eclipses and Occultation.

Concepts of geodesy and surveying, Earth's gravity field and the geoid, and measurement techniques applied to geomatics are examined. Field studies include the use of the level, the total station, and GPS for doing distance and angle measurements, leveling, traversing and topographic surveying. Principles of global positioning system (GPS) and its applications.

Labs:

- Laboratory Sessions will be held in the ISPA observatory for Observational Astronomy.
- GPS for doing distance and angle measurements, leveling, traversing and topographic surveying.

Books Recommended:

- Ball R S (2011), A treatise on Spherical Astronomy, Nabu Press.
- Greene D (2010), An introduction to Spherical and Practical Astronomy, Nabu Press.
- Anderson J M and Mikhail E M (1998), Surveying Theory and Practice, Seventh Edition, MCB/McGraw-Hill.
- Wolf P R and Ghilani C D (2005), Elementary Surveying – An Introduction to Geomatics, Eleventh Edition, Prentice Hall.
- Kavanagh B F (2006), Surveying: Principles and Applications, Seventh Edition, Prentice Hall.
- Taff L G (1991), Computational Spherical Astronomy, Krieger Publication Co.
- Chauvenet W (2007), A Manual of Spherical And Practical Astronomy V2: Theory And Use of Astronomical Instruments, Kessinger Publishing, LLC.
- Brunnnow F (2011), Spherical Astronomy (Classic Reprint), Forgotten Books.
- Main R (2010), Practical and Spherical Astronomy, Nabu Press.
- Green R M (1985), Spherical Astronomy, Cambridge University Press.
- Smart W M and Green R M (1977), Textbook on Spherical Astronomy, Sixth Edition, Cambridge University Press.

SPA 503: Mathematical Physics (3+0)

Methods of solution reduction of order, undetermined coefficients, variation of parameter and Green's function. Power series solution. Legendre and Bessel's equations. Properties of Legendre polynomials and Bessel's functions. Laplace integral transforms, solving initial value problem using Laplace transform, Fourier series, Fourier transform. Partial differential equations of Mathematical Physics. Method of separation of variable. Boundary value problems relating vibration of strings and membranes, heat conduction and potential theory.

Curve theory, Frenet equations, Riemannian Manifold, Tensors, Covariant derivative, Curvature Tensor, Geodesics.

Books Recommended:

- Kraut E A (2007), Fundamentals of Mathematical Physics, Dover Publications.
- Kusse B R and Westwig E A (2006), Mathematical Physics: Applied Mathematics for Scientists and Engineers, Second Edition, Wiley-VCH.

- Kirkwood J (2012), Mathematical Physics with Partial Differential Equations, Academic Press.
- Brown J and Churchill R (2011), Fourier Series and Boundary Value Problems, Eighth Edition, McGraw-Hill Science/Engineering/Math.
- Rabenstein A L (1992), Elementary Differential Equations with Linear Algebra, Fourth Edition, Brooks Cole.
- Boyce W E and DiPrima R C (2008), Elementary Differential Equations and Boundary Value Problems, Ninth Edition, Wiley, New York.
- O'Neil P V (2008), Beginning Partial Differential Equations, Second Edition, Wiley Interscience Pub., John Wiley and Sons Inc.
- Rainville E D, Bedient P E and Rich and Bedient E (1996), Elementary Differential Equations, Eighth Edition, Prentice Hall.
- Presley A (2010), Elementary Differential Geometry; Springer.
- Kunel W (2005), Differential Geometry: Curves-Surfaces-Manifold, Second Edition, Student Mathematical Library

SPA 505: Introduction to Database and Computer Programming (2+1)

Databases, and their typical users, DBMS concepts, terminology, and architecture. concepts of the Entity-Relationship (ER) model, ER diagrams, data abstraction and semantic data modeling concepts, EER diagrams, databases design and implementation, conceptual modeling techniques used in database systems, relational data model, its integrity constraints and update operations, and the operations of the relational algebra, relational DBMSs, introduction to object databases, comprehensive introduction to object-relational systems.

Introduction to operating System, Exploring print, C building blocks, variables, input/output, operators, loops: for loops, while loop, do while loop, decisions, if statement, if else statement, else if construct switch statement, conditional statement, function, function that return a value using argument to pass data to a function, external variable, arrays and strings, pointers, structure, files and introduction to C++ concepts.

Recommended Books:

- Elmasri R and Navathe S B (2010), Fundamentals of Database Systems, Sixth Edition, Addison Wesley.
- Hoffer J A, Venkataraman R and Topi H, (2010), Modern Database Management, Tenth Edition, Prentice Hall.
- Coronel C, Morris S and Rob P, (2009), Database Systems: Design, Implementation and Management, Ninth Edition, Course Technology.
- Roy P V and Haridi S (2004), Concepts, Techniques and Models of Computer Programming, The MIT Press.
- Appleman D and Ishida S (2000), How Computer Programming Works, Apress.
- Lafore R, (1990), The Waite Group's C Programming Using Turbo C++, Second Edition, Sams Publishing.
- Schildt H (1990), Turbo C/C++, Osborne/McGraw-Hill,U.S.
- Bloom E P (1991), The Turbo C++ Trilogy, Windcrest.
- Sipser M (2005), Introduction to the Theory of Computation, Second Edition, Course Technology.

SPA 507: Foundation of Remote Sensing (2+1)

Introduction to Remote Sensing, Remote Sensing and its Physical Principles, Electromagnetic Radiation, EM Spectrum, Interaction of Electromagnetic Radiation with Atmosphere and with the Earth's Surface, Atmospheric Windows, Signatures in Remote Sensing, Significance of Multi Spectral Imagery, Resolutions and its meanings (Spectral, Spatial, Temporal and Radiometric), Colors and Human Vision, Color Models, Types of Photographic Films, Sensors and their types, optical, microwave, hyper spectral and ultraspectral sensors, imaging systems (RBV, MSS, TM, HRV, HRPT/APT/AVHRR, ETM, HRV, QuickBird, GeoEye, EO-1 etc.) non-imaging systems (Radarsat, SAR, SIR, etc.), satellites: types and functions, unmanned satellites (generations, meteorological satellites, earth resources satellites, telecommunication satellites, spy satellites, scientific satellites) manned satellites, space shuttles, platforms, ground receiving stations and reception of data, Data Acquisition (Ground Receiving Stations), Image Processing (Continuous and Discrete), Image Interpretation and Analysis.

Labs:

Introduction to labs, single band image interpretation, false color predictions, false color composite images interpretation, visual interpretation of aerial photographs, various sensors data comparison, thermal infrared image interpretation, introduction to ERDAS imagine, display, geo-linking, zooming, identification of targets.

Books Recommended:

- Campbell J B and Wynne R H (2011), Introduction to Remote Sensing, Fifth Edition, The Guilford Press, New York.
- Sabins F F, (2007), Remote Sensing: Principles and Interpretation, Third Edition, Waveland Press Inc.
- Gibson P J (2000), Introductory Remote Sensing: Principles and Concepts, Routledge, USA.
- Jensen J (2006), Remote Sensing of the Environment: An Earth Resources Perspective, Prentice Hall, London.
- Lillesand T, Kiefer R W and Chipman J (2007), Remote Sensing and Image Interpretation, Sixth Edition, John Wiley, USA.
- Schowengerdt R A (2006), Remote Sensing: Models and Methods for Image Processing, Third Edition, Academic Press.

SPA 509: Statistical Methods (2+1)

Review of Probability; element of probability, properties of probability, Empirical Distributions and Exploratory Data Analysis; Stem-and-Leaf display, Boxplots, Histograms, Kernel Density Smoothing, Cumulative Frequency Distribution, Power Transformation, Standardized Anomalies. Pearson Correlation, Correlation Matrix for Higher Dimensional Data, Parametric Probability Distributions; Discrete and Continuous Distributions (Binomial, Poisson, Gaussian, Gamma, Extreme value Distributions), Goodness of Fit test, Parametric Fitting using Maximum likelihood, Statistical Simulation. Hypothesis Testing. Statistical Forecasting; Linear Regression, Nonlinear Regression, Ensemble Forecasting. Time Series; Markov Chains, First Order Autoregression, AR (2) Model, Autoregressive Moving Average Models. Frequency Analysis. Multivariate Analysis; Principal Component Analysis

Labs:

Analysis and modeling of astrophysical data using statistical packages (e.g. STATISTICA, MINITAB, SPSS, SAS, R, MINITAB, etc.)

Recommended Books:

- Wilks D.\ S (2006), Statistical Methods in the Atmospheric Sciences, Elsevier.
- Freund R J, Mohr D and Wilson W J (2010), Statistical Methods, Third Edition, Academic Press.
- Thomas J B (2009), An Introduction to Applied Probability and Random Processes, Tenth Edition, Wiley, New York.
- Ross S M (2009), Introduction to Probability Models, Tenth Edition, Academic, New York.
- Wilcox R R (2012), Introduction to Robust Estimation and Hypothesis Testing, Third Edition, Academic Press.
- Cowan G (1998), Statistical Data Analysis, Clarendon, Oxford.
- Stirzaker D (1994), Elementary Probability, University Press, Cambridge.
- Krzanowski W J (1998), Statistical Data Analysis, Arnold, London.
- Stark H and Woods J W (1994), Probability, Random Processes and Estimation Theory for Engineers, Second Edition, Prentice Hall.

COURSE OUTLINES FOR SECOND SEMESTER

SPA 502: Astronomy (2+1)

Solar system, Planets and minor objects, Planetary data, Nature of radiations from cosmos, Interaction of light with matter, Blackbody Radiations, Telescopes and their types, Telescopes function, Data gathering and handling, H-R diagrams, Dwarf Stars, Red Giant Stars, Supergiant Stars, Brown Dwarf, Nebulae, Formations of Stars in Nebula, Protostars, Stellar structure and evolution, pre-main-sequence and main-sequence stars, Sources of stellar energy, The Sun and solar neutrino puzzle, Stellar magnitudes, Colors and temperatures, Binary stars, Visual, spectroscopic and eclipsing binaries, Variable stars, Novae, Supernovae, Compact stars, White dwarfs and neutron stars, Pulsars, quasars, galaxies and their types, Active galaxies.

Labs:

- Student will use and hands-on experience on Red Shift & Starry Night Pro Software.
- Students will study the techniques of astronomical data acquisition and give them some hands-on experience with a telescope- from simply being able to find and view a certain object in the sky, to taking photographs, CCD images, photometry and spectroscopic data.
- Telescopes - Different types of telescopes - Angular resolution and Diffraction Limited Resolution - Image formation in a camera - Plate Scale - Observatories (Ground Based & Space Based) - Focal Plane Instruments-Imagers - Photometers - Spectrometers - CCDs and their use in astronomy - Detectors for Optical, Infrared, UV, X-rays, and Gamma-rays - Effect of Atmosphere.

Recommended Books:

- Seeds M A. and Backman D (2010), Foundation of Astronomy, Eleventh Edition, Brooks Cole.
- Cox A N (2001), Astrophysical Quantities, Fourth Edition, Springer.
- Seeds M A and Backman D (2012), The Solar System, Eighth Edition, Brooks Cole.
- Padmanabhan T (2001), Theoretical Astrophysics: Volume 2, Stars and Stellar Systems, Cambridge University Press.
- Chaisson E and McMillan S (2010), Astronomy Today, Seventh Edition, Benjamin Cummings.
- Chaisson E and McMillan S (2009), Astronomy: A Beginner's Guide to the Universe, Sixth Edition, Benjamin Cummings.
- Arny T and Schneider S (2009), Explorations: Introduction to Astronomy, Sixth Edition, McGraw-Hill Science/Engineering/Math.
- George C (2010), A text-book of Astronomy, Nabu Press.
- Hester J, Smith B, Blumenthal G, Kay L and Voss H (2010), 21st Century Astronomy, Third Edition, W. W. Norton & Company.

SPA 504: Classical Mechanics (3+0)

Elementary Principles: Brief survey of Newtonian mechanics of a system of particles, constraints, D'Alembert's principle. Lagrange's equation and its applications. Variational Principles: Calculus of variation and Hamilton's principle, Derivation of Lagrange's equation from Hamilton's principle. Two body Central Force Problem: Two body problem and its reduction to one body problem, Equation of motion and solution for one body problem, Kepler's laws laboratory and centre of mass system, Rutherford scattering. Rigid Body Equation of Motion: Angular momentum, moment of inertia tensor, Rigid body problems and Euler's equations. Hamilton Equation of Motion: Legendre transformation and Hamilton equations of motion, Conservation theorems, principles of least actions. Canonical

Transformations: Examples of canonical transformations, Lagrange and Poisson brackets, Liouville's theorem.

Recommended Books:

- Goldstein, H, Poole C P and Safko J L (2001), Classical Mechanics, Third Edition, Addison Wesley.
- Johnson P W (2010), Classical Mechanics with Applications, World Scientific Publishing Company.
- Meirovitch L (2010), Methods of Analytical Dynamics, Dover Publications.
- Thornton S T and Marion J B (2003), Classical Dynamics of Particles and Systems, Fifth Edition, Brooks Cole.
- Corben H C and Stehle P (1994), Classical Dynamics, Second Edition, Dover Publications.
- Caratheodory C and Carathéodory C (1999), Calculus of Variations and Partial Differential Equations of First Order, Third Edition, American Mathematical Society.
- Taylor E F and Wheeler J A (1992), Spacetime Physics, Second Edition, W. H. Freeman.
- Meirovitch L (2007), Methods of Analytical Dynamics, McGraw Hill, New York.
- Taylor J R (2005), Classical Mechanics, University Science Books.
- Morin D (2008), Introduction to Classical Mechanics: With Problems and Solutions, Cambridge University Press.
- Gregory R D (2006), Classical Mechanics, Cambridge University Press.
- Kibble T W B and Berkshire F H (2004), Classical Mechanics, Fifth Edition, World Scientific Publishing Company.
- Sheck, F (1988), Mechanics, Springer Verlag, Berlin.

SPA 506: Numerical Computing (2+1)

Error analysis, Round-Off Error; Absolute, Relative and Percentage Error; Significant Digits; Propagated Error; Error in Original Data, etc. Truncation Errors.

Solution of Nonlinear Equations, Approximation of Functions, Bisection Method, Linear Interpolation, Newton's Method, Quasi-Linearization. Solving System of Linear Equations, Gaussian Elimination, Crout's Decomposition Scheme, QD-Algorithm, Efficient Computation and Use of Determinants; Matrix Inversion; Iterative Methods. System of Nonlinear Equations.

Interpolation: Interpolating Polynomials; Extrapolation; Divided Differences and Finite Differences of Various Orders, Difference Tables, Forward, Backward and Central Differences. Symbolic Derivatives. Interpolation with Unequal Intervals. Inverse Interpolation. Curve Fitting by Various Methods: Splines, Cubic Spline Curves; Least Squares Approximation.

Numerical Differentiation, Higher Order Numerical Derivatives. Numerical Integration, Simpson's Rule. Numerical Solution of Ordinary Differential Equations: Initial Value Problem; Solving Ordinary Differential Equations, Various Order Runge-Kutta Methods, Boundary Value Problems.

Labs:

Solving and implementing Numerical Algorithms using MATLAB.

Recommended Books:

- Burden R L and Faires J D (2010), Numerical Analysis, Ninth Edition, Brooks Cole, Boston.
- Heinbockel J H (2006), Numerical Methods for Scientific Computing, Trafford Publishing.
- Gerald C F and Wheatley P O (2003), Applied Numerical Analysis, Seventh Edition, Addison-Wesley., Mass. Addison Wesley;

- Palm W (2004), Introduction to MATLAB 7 for Engineers, Second Edition, McGraw-Hill Science/Engineering/Math.
- Trauth M H (2010), MATLAB Recipes for Earth Sciences, Third Edition, Springer.
- Moler C B (2008), Numerical Computing with MATLAB, Revised Reprint Edition, Society for Industrial and Applied Mathematics.
- Cheney E W and Kincaid D R (2007), Numerical Mathematics and Computing, Sixth Edition, Brooks Cole.
- Atkinson K E (1989), An Introduction to Numerical Analysis, Wiley, New York.
- Gilat A (2010), MATLAB: An Introduction with Applications, Fourth Edition, Wiley.
- Hein J L (1996), Theory of Computation: An Introduction, Jones & Bartlett Pub.
- Trauth M H (2007), MATLAB Recipes for Earth Sciences

SPA 508: Geographical Information Science (2+1)

Introduction and overview of GIS: introduction, definitions, components, functional sub-science. Evolution and application of GIS, data models (raster data model, vector data model, attribute data model), data acquisition techniques (data sources, map and data sources in Pakistan, data capturing techniques and procedures), data transformation, visualization of spatial data (layers and projections), map design: (symbols to portray points, lines and volumes, graphic variables, visual hierarchy), data classification graphic approach, mathematical approach spatial relationships (topology), spatial analysis: overlay analysis, spatial analysis, neighborhood functions, network and overlay analysis, buffering, spatial data quality: components of data quality, micro level, components, macro level components, usage components, sources of error, and accuracy. Digital Surface Modeling in GIS, Applications of GIS for Land Resource Management, Regional Planning and Land Use Change Analysis, Errors and Uncertainty, Global Positional System (GPS), Future of GIS, GIS in Pakistan (Data Sources and Applications)

Labs:

The lab, exercises based on hands-on desktop GIS state-of-art software (MapInfo and ArcGIS) will provide extensive training in data collection, database development and image processing for any real world example, following are the activities involved: introduction to GIS lab. (hardware/software), introduction to GIS software, raster/vector/attribute data display, scanning, digitization, co-ordinate based point mapping, raster/vector conversion, data layer integration and display of different projections, map layout, data classification and thematic mapping, handling with topological errors, overlay analysis, network analysis.

Recommended Books:

- Arctur D K (2004), Designing Geo databases: Case Studies in GIS Data Modeling, ESRI Press, California.
- Aronoff S (2009), Geographic Information Sciences: A Management Perspective, Sixth Edition, WDL Publications, Ottawa, Canada.
- Chang K T (2010), Introduction to GIS, Second Edition, McGraw-Hill & Tata, New Delhi.
- Clarke K (2004), Getting started with Geographic Information Science, Second Edition, Prentice Hall, New Jersey.
- Cornelius S. and Carver. S (2003), An introduction to Geographic Information Science, Second Edition, Addison Wesley, New York.
- Jankowski P and Nyerges T (2001), Geographic Information Sciences for Group Decision Making, Taylor & Francis.
- Heywood I, Cornelius S and Carver S (2011), An Introduction to Geographical Information Systems, Fourth Edition, Prentice Hall.

- Langley P A, Goodchild, M F and Rhind D. W., (2005), Geographic Information Science: Principles and Applications, Second Edition, ESRI Press, California.
- Longley P A (2003) Advanced Spatial Analysis, ESRI Press, California.
- DeMers M N (2008), Fundamentals of Geographical Information Systems, Fourth Edition, Wiley.
- Narayan Panigrahi, (2009), Geographical Information Science, CRC Press.
- Jensen J R (1998), Manual of Remote Sensing, Principles and Applications of Imaging Radar, Third Edition, Wiley, USA.

SPA 510: Atmospheric Science (3+0)

Components of Earth System, Hydrologic Cycle, Carbon Cycle, Oxygen in the Earth System.

Atmospheric Thermodynamics; Gas Laws, Hydrostatic Equation, First Law of Thermodynamics, Adiabatic Process, Water Vapor in Air, Static Stability, Second Law of Thermodynamics and Entropy.

Radiative Transfer; Spectrum of Radiation, Quantitative Description of Radiation, Blackbody Radiation, Physics of Scattering and Absorption, Radiative Transfer in Planetary Atmospheres.

Atmospheric Chemistry; Composition of Troposphere Air, Sources, Transport and Sink of Trace Gases, Tropospheric Trace Gases, Tropospheric Aerosol, Air Pollution, Tropospheric Chemical Cycles and Stratospheric Chemistry.

Cloud Microphysics; Nucleation of Water Vapor Condensation, Microstructure of Warm Cloud, Growth of Cloud Droplets in Warm Clouds, Microphysics of Cold Clouds.

Recommended Books:

- Wallace J and Hobbs P (2006), Atmospheric Sciences: An Introductory Survey, Elsevier.
- Frederick J E (2008), Principles of Atmospheric Sciences, Jones and Bartlett Publisher.
- Holton J R (2004), An Introduction to Dynamics Meteorology, Elsevier.
- Wallace J H and Hobbs P V (2006), Atmospheric Science, Second Edition: An Introductory Survey, Second Edition, Academic Press.
- Frederick J E (2008), Principles of Atmospheric Science, Jones & Bartlett Publisher.
- Hewitt C N et al. (2003), Handbook of Atmospheric Science: Principles and Applications, Wiley-Blackwell
- Moran J M (2006), Weather Studies: Introduction to Atmospheric Science, Third Edition, AMS Online.

COURSE OUTLINES FOR THIRD SEMESTER

SPA 601: Atmospheric Dynamics (3+0)

Momentum Equation, Scale Analysis of Equation of Motion, Continuity Equation, Thermodynamics Energy Equation, Basic Equation in Isobaric Coordinates, Vertical Motion, Circulation and Vorticity, Planetary Boundary Layer, Quasi-Geostrophic Analysis, Atmospheric Waves, Baroclinic Instability, Mesoscale Circulation and General Circulation.

Recommended Books:

- Mak M (2011), Atmospheric Dynamics, Cambridge University Press.
- Holton J R (2004), An Introduction to Dynamics Meteorology, Elsevier.
- Lynch A H and Cassano J J (2006), Applied Atmospheric Dynamics, Wiley.
- Martin J E (2006), Mid-Latitude Atmospheric Dynamics: A First Course, Wiley.
- Green J (2004), Atmospheric Dynamics, Cambridge University Press.
- Szunyogh I (2010), Applicable Atmospheric Dynamics: Techniques for the Exploration of Atmospheric Dynamics, World Scientific Pub. Co. Inc.

SPA 603: Space Flight Dynamics (3+0)

Coordinate Systems and Rotation Matrix, Euler Axes and Principle Angle, Euler Angles, Particle Kinematic in a Moving Frame, variable Mass Bodies, Rotation and Translation of a Body, n-Body Problem, Two Body Problem; Geometry of Two Body Trajectories, Lagrange's coefficients, Kepler's Equation for Elliptical Orbit, Position and Velocity in a Hyperbolic Trajectory, Parabolic Escape Trajectory. Celestial Frame and Orbit Elements; Orbit Determination. Rocket Propulsion; Rocket Engine, Rocket Equation and Staging, Optimal Rocket. Planetary Atmosphere; Hydrostatic Equilibrium. Element of Aerodynamics; Aerodynamics Force and Moment, Fluid dynamics, Flow regime, Continuum Flow, Continuum Viscous Flow and Boundary Layer, Rarefied Flow. Air breathing Propulsion; Ideal Momentum Theory, Propeller Engine, Jet Engine.

Recommended Books:

- Tewari A (2007), Atmospheric and Space Flight Dynamics, Birkhäuser Boston.
- Cook M (2007), Flight Dynamics Principles, A Linear Systems Approach to Aircraft Stability and Control, Second Edition, Butterworth-Heinemann.
- Etkin B (2005), Dynamics of Atmospheric Flight, Dover Publications.
- Stengel R F (2004), Flight Dynamics, Princeton University Press.
- Hughes P C (2004), Spacecraft Attitude Dynamics, Dover Publications.
- Roskam J (2003), Airplane Flight Dynamics and Automatic Flight Controls: Part 2, Darcorporation.
- Sivaram C (2009), Rocket Dynamics and Space Flight, ANE Books.
- William E. Wiesel., (1996), Spaceflight Dynamics, Second Edition, McGraw-Hill.
- Etkin B and Reid L D (1995), Dynamics of Flight: Stability and Control, Third Edition, Wiley.
- Schmidt L V (1998), Introduction to Aircraft Flight Dynamics, American Institute of Aeronautics & Ast. (AIAA) Education Series.

SPA 605: Quantum Mechanics (3+0)

Origin of Quantum theory, Foundation of wave mechanics, Heisenberg Uncertainty principle, Waves and particles, and wave function, Probability density and probability current density, Orthogonal and Orthonormal Wave functions, Operator Algebra, Schrödinger. Wave equation, Wave function of a free particle. Time dependent and Time independent Schrödinger Wave equation, Applications of Schrödinger

Wave equation, Potential wells, potential barrier, and tunneling of particles, Harmonic Oscillator and rigid rotators, Hydrogen Atom, Angular Momentum and Spin of particles, Pauli Exclusion Principle, Wave functions of fermions and bosons, time independent and dependent perturbation theory.

Recommended Books:

- Sakurai J J and Napolitano J J (2010), Modern Quantum Mechanics, Second Edition, Addison Wesley.
- David A. B. Miller, (2008), Quantum Mechanics for Scientists and Engineers, Cambridge University Press.
- Levi A F J (2003) Applied Quantum Mechanics, Cambridge University Press, UK.
- Randy H (2000), Non-classical Physics, Addison Wesley, New York.
- Sinha S P (2001), Quantum Mechanics, Tata McGraw Hills, New Delhi.
- Griffiths D J (2004), Introduction to Quantum Mechanics, Second Edition, Benjamin Cummings.
- Townsend J (2000), A Modern Approach to Quantum Mechanics, Second Edition, University Science Books.
- Zettili N (2009), Quantum Mechanics: Concepts and Applications, Second Edition, Wiley.
- Liboff R (2002), Introductory Quantum Mechanics, Fourth Edition, Addison Wesley.
- Reed B C (2007), Quantum Mechanics, Jones & Bartlett Publisher.
- Bransden B H and Joachain C J (2000), Quantum Mechanics, Second Edition, Benjamin Cummings.
- Gottfried K and Yan T M, (2004), Quantum Mechanics: Fundamentals, Second Edition, Springer.

Course Outlines for Optional Subject-I & II

SPA 607: Cosmology (3+0)

Friedman's equations. The cosmological constant. Model universes. Scaling laws. Equations of state. The benchmark universe. Nuclear physics and nucleosynthesis. Neutrinos and other elementary particles. Inflation. The flatness problem and the fine tuning problem. The cosmic microwave background and its fluctuations. The accelerating universe. The concordant picture.

Recommended Books:

- Duncan T and Tyler C (2008), Your Cosmic Context: An Introduction to Modern Cosmology, Benjamin Cummings.
- Cheng T P (2005), Relativity, Gravitation, and Cosmology, Oxford.
- Gron O and Sigbjorn Hervik, (2007), Einstein's General Theory of Relativity: With Modern Applications in Cosmology, Springer.
- Carmeli M (2006), Cosmological Relativity: The Special and General Theories of the Structure of the Universe, World Scientific Publishing Company.
- Plebanski J and Andrzej Krasinski, (2006), An Introduction to General Relativity and Cosmology, Cambridge University Press.
- Raychaudhuri A K, Banerji S and Banerjee A (2003), General Relativity, Astrophysics, and Cosmology, Second Edition, Springer.
- Bennett J O, Donahue M, Schneider N O and Voit M (2009), The Cosmic Perspective: Stars, Galaxies, and Cosmology, Sixth Edition, Addison-Wesley.
- Ryden B (2002), Introduction to Cosmology, Benjamin Cummings.
- Harrison E (2000), Cosmology: The Science of the Universe, Second Edition, Cambridge University Press.
- Islam J. N., (1992), An Introduction to Mathematical Cosmology, Cambridge.
- Bertotti B, Balbinot R, Bergia S and Messina A (1991), Modern Cosmology in Retrospect, Cambridge University Press.

SPA 609: Statistical Physics and Space Plasma (3+0)

Statistical probability, canonical and micro canonical ensemble, partition function, Entropy, equipartition theorem, free energy, Statistical Distribution and Mean values: Mean free path and microscopic calculations of mean free path. Distribution of Molecular speeds, Distribution of energies: Maxwell distribution; Maxwell-Boltzmann energy distribution; Internal energy of an ideal gas. Brownian motion, Qualitative description. Diffusion, Conduction and Viscosity. Liouville's equation
Introduction to Plasmas, Single-Particle Motions, Plasmas as Fluids, Elementary Plasma Waves, Diffusion and Resistivity, Dusty Plasmas.

Recommended Books:

- Schindler K (2010), Physics of Space Plasma Activity, Cambridge University Press.
- Bittencourt J A (2004), Fundamentals of Plasma Physics, Third Edition, Springer-Verlag New York.
- Goldston R J and Rutherford P H (1995), Introduction to Plasma Physics, IOP Publishing Ltd.
- Reif F (2008), Fundamentals of Statistical and Thermal Physics, Waveland Press Inc.
- Cravens T E (2004), Physics of Solar System Plasmas, Cambridge University Press.
- Parks G (2003), Physics Of Space Plasmas: An Introduction, Second Edition, Westview Press.
- Somov B V (2000), Cosmic Plasma Physics, Springer.

- Kardar M (2007), Statistical Physics of Particles, Cambridge University Press.
- Reichl L E (2009), A Modern Course in Statistical Physics, Wiley-VCH.
- Kittel C (2004), Elementary Statistical Physics, Dover Publications.
- Krapivsky P L, Redner S and Ben-Naim E (2010), A Kinetic View of Statistical Physics, Cambridge University Press.
- Huang K (2009), Introduction to Statistical Physics, Second Edition, Chapman and Hall/CRC.
- Wannier G H (2010), Statistical Physics, Dover Publications.
- Kadanoff L P (2000), Statistical Physics: Statics, Dynamics and Renormalization, World Scientific Pub. Co. Inc.
- Chen F F (1984), Introduction to Plasma Physics and controlled fusion, Vol. 1, Second Edition, Springer.

SPA 611: Global Positioning Systems (2+1)

Introduction to GPS, Satellite Navigation System, Space Segment, Control Segment, User Segment, GPS Positioning Services Specified In the Federal Radio, Navigation Plan, Precise Positioning Service, Standard Positioning Service, GPS Data, Position and Time from GPS, Code Phase and Pseudo-Range Navigation, Receiver Position, Velocity, and Time, Carrier Phase Tracking (Surveying), GPS Satellite Signals, GPS Error Sources, Differential GPS Techniques, Differential Code-Phase Navigation, Differential Carrier-Phase Surveying, Common-Mode Time Transfer, GPS Techniques and Project Costs, Exploration of Advance System as Differential GPS, Applications in Pakistan: Tracking and Car Navigation

Labs:

GPS value reading, Easting Northing & elevation, Map Projections and Datum Settings, GPS based surveys, GPS coordinates uploading onto GIS System, tracking and data processing, Planimetric & vertical errors calculations, GPS Project.

Recommended Books:

- Kennedy, M (2010), The Global Positioning System and GIS: An Introduction, Third Edition, Taylor & Francis, New York.
- Zarchan, P. (2006), Global Positioning System: Theory and Application, Volume I, American Institute of Aeronautics and Astronautics, Inc., Washington DC.
- Heywood I, Cornelius S. and Carver S (1999), An introduction to Geographic Information System, Addison Wesley Longman, New York, second edition.
- Aronoff S (1995), Geographic Information Systems: A Management Perspective, WDL Publications, Ottawa, Canada, Forth edition.
- GPSCO (1992), Getting started with GPS Surveying, GPSCO Land Information centre, NSW, Australia.
- El-Rabbany A (2006), Introduction to GPS: The Global Positioning System, Second Edition, Artech House Publishers.
- Misra P and Per Enge P (2010), Global Positioning System: Signals, Measurements, and Performance, Second Edition, Ganga-Jamuna Press.
- Kennedy M (2009), The Global Positioning System and ArcGIS, Third Edition, CRC Press.
- Grewal M S, Weill L R and Andrews A P (2007), Global Positioning Systems, Inertial Navigation, and Integration Second Edition, Wiley-Interscience.

SPA 613: Navigation & Space Instrumentation (3+0)

Space navigation including inertial and celestial navigation. Gyroscope, accelerometers, GPS, Worldwide Navigational Systems. Principles of guidance systems for spacecraft, launch vehicles, homing and

ballistic missiles. Optimal guidance, Interplanetary transfer guidance with low thrust, Principles of inertial navigation, theory and applications of the Global Positioning System.

Process of measurement standards and dimensional units of measurement, sensors, counters, displacement and dimensional measurements, and stress & strain measurement analysis, fluid flow measurement, temperature measurement and motion measurement. Experimental portion includes use of measuring instruments like autocollimator, screw pitch measuring machine, tolerator, horizontal metro scope, interference microscope, interferometer, surface roundness testing machine and Mechanical & Electronic comparators.

Recommended Books:

- Hofmann-Wellenhof B, Lichtenegger H and Wasle E (2007), GNSS - Global Navigation Satellite Systems: GPS, GLONASS, Galileo and more, Springer.
- Tyler A (1996), Celestial Navigation, Ballantine Books.
- Groves P D (2007), Principles of GNSS, Inertial, and Multi-Sensor Integrated Navigation Systems, Artech House.
- Emadzadeh A A and Speyer J L (2011), Navigation in Space by X-ray Pulsars, Springer.
- Rycroft M J (2003), Satellite Navigation Systems: Policy, Commercial and Technical Interaction, Springer.
- Bezerra E (2010), Reconfigurable Systems in Space Instrumentation: Moving from Microprocessors to FPGAs, LAP Lambert Academic Publishing.
- Dally J W, Riley W F and McConnell K G, (1993), Instrumentation for Engineering Measurements, Second Edition Wiley.
- De Silva C W (2007), Vibration Monitoring, Testing, and Instrumentation, CRC Press.
- Instrumentation in the Aerospace Industry, (1982), volume 28, Instrument Society of America.

SPA 615: Space and Satellite Communication (3+0)

Transmission channels, Linear modulation, Exponential modulation, Noise in analogue carrier wave transmission, Carrier wave systems, Base-band digital transmission, Sampling and pulse coded modulation, Band-pass digital transmission, Error control coding, Information theory, Advanced system applications: Sampling, interpolation, and aliasing, Definition and Role of layered approach, SI and TCP/IP models, Physical layer: Digital transmission Systems.

Principles of Satellite Communication, Electromagnetic Spectrum, Interactions of Microwaves with Atmosphere, Communication Satellite System and Sub-Systems (Attitude and orbit control system), Telemetry, Tracking and Command (TT&C) Power Sub-System, Communication Sub-Systems, Spacecraft Antenna, Satellite Link Design (Basic transmission theory, RF Links, Optimization of RF Link, System Noise Temperature, Noise Figure, G/T Ratio, Down/Uplink Design and Complete Link Budget for LEO, MEO & GEO Satellites), Earth Station Parameters, Base Band Formation, Multiple Access Techniques, Post-launch space operations, Virtual Ground Station Development, Special Purpose Communication Satellites (DBS, INMARSAT, VSAT, SARSAT and LEO Satellites etc.).

Recommended Books:

- Maral G and Bousquet M (2010), Satellite Communications Systems: Systems, Techniques & Technology, Fifth Edition, John Wiley & Sons.
- Roddy D (2006), Satellite Communications, Fourth Edition, McGraw-Hill Professional.
- Pratt T, Bostian C and Allnutt J (2003), Satellite Communications, Second Edition, John Wiley & Sons.
- Agarwal D C (2002), Satellite Communications, Fifth Edition, Khanna Publishers, India.
- Ackroyd B (1990), World Satellite Communication Systems: An Earth Station Design, CRC Press.

- Balanis C A (2005), Antenna Theory: Analysis & Design, Third Edition, Wiley-Interscience.
- Elbert B, Lee S.W. and Lo. Y. T., (2001), Satellite Communication Ground Segment and Earth Station, John Wiley, New York.
- Enrico Del Re and Ruggieri M (2007), Satellite Communications and Navigation Systems, Springer.
- Corazza G E (2007), Digital Satellite Communications, Springer.
- Fares A (2007), Satellite Communications Engineering, BookSurge Publishing.
- Swiston M (2007), Satellite Communication Systems, Exposure Publishing, an imprint of Meadow Books.
- Martin D H, Anderson P R and Bartamian L (2007), Communication Satellites, AIAA.
- Frenzel K., (1999), Communication Electronics, Third Edition, McGraw-Hill, New York.
- Calcutt D and Tetley L (1994), Satellite Communication, Principles & Applications, Butterworth-Heinemann.

SPA 617: Fundamentals of Physical Oceanography (2+1)

Historical Review of Oceanography, geological background, physical properties of sea water, Density, Salinity, Temperature, light and sound in the ocean, sound channel, shadow zones, thermal and mechanical properties of sea water, Equation of state for sea water, properties of sea ice, Physical laws and Classification of Forces and motion in the sea, conservation of mass, Equation of motion, role of non linear terms in the equation of motion, Waves and Tides, Shallow Water and Deep Water Waves, Wave Reflection and Refraction, Tides Origins and Characteristics, Tidal forcing, Tides near coasts, tidal mixing in Estuaries, River outflow plumes, Fronts, Tidal currents, Response of the Upper Ocean to Winds, Surface currents, Ekman Layer and Ekman Mass Transport, upwelling, geostrophic flow, hydrostatic equilibrium, Inertial flow, vorticity, westward intensification, Gulf Stream, Coastal Process, Longshore currents, Rip currents, Swell, Estuarine circulation, Mass Transport in the Ocean, Water Masses, water mass identification and formation, distribution of various properties, T-S relationship, water mass formation in the North Indian Ocean, thermohaline effects, deep ocean circulation, Ocean Atmosphere Heat Exchange, Monsoon circulation, El Nino-Southern Oscillation, Coastal Hazards, Cyclones, storm Surges, High Waves, Coastal Flooding, Salt Water Intrusion, Tsunami, Tsunami Early Warning System for Indian Ocean rim countries

Labs:

- Introduction of Oceanographic and Meteorological Instruments, CTD, Current meters, tide gauges, wave rider buoys, Eco sounder, Nansen and Niskin bottles, Multi Rosette, reversing thermometer, anemometer, barometer etc.
- Methods for measurements of oceanographic and marine meteorological parameters, measurements of physical parameters, tides, currents, waves, temperature, salinity etc.
- Sampling procedures for collections of sediments, sea water and marine fauna & flora.
- Analysis of oceanographic and meteorological data, plotting of currents, Temperature and salinity profiles, T-S diagrams, tidal charts.
- Provision of observing platform: Research vessels, mooring, floats and drifters.

Recommended Books:

- Stewart, R. H. (2007). Introduction to Physical Oceanography, Texas A & M University.

- Emery, W. J. and Thomson, R. E. (2001). Data Analysis Methods in Physical Oceanography, Elsevier.
- Pickard, G. L., Emery, W. J. (1990). Descriptive Physical Oceanography, Fifth Edition, Butterworth-Heinemann.
- Perry, H. (1977). The ocean atmosphere system, Longman- London and New York.
- Csanady, G. T. (2001). Air-Sea Interaction, Cambridge University Press.
- Sverdrup, H.U., Johnson, M.W. and Fleming, R.W. (1966). The Oceans: Their Physics, Chemistry and General Biology. Prentice-Hall, Englewood, NJ.
- Instruction manual for Obtaining Oceanographic Data (1975), NAVAL OCEANOGRAPHIC OFFICE WASHINGTON.
- W. Duing, P. Duncan, J. Meincke (1971). Study Questions in Physical Oceanography, University of Miami, special Publication.
- John A. Knauss (2005), Introduction to Physical Oceanography, Waveland Pr Inc; 2 edition.
- Oba D (2011). Ocean tide, Author House Publishing.
- Knauss J A (2005), Introduction to Physical Oceanography, Second Edition, Waveland Pr. Inc.
- Nelson C B and Hardee M (2005), Introduction to Oceanography Lab Manual, Kendall-Hunt.
- Denny M (2008), How the Ocean Works: An Introduction to Oceanography, Princeton University Press.
- Oba D (2011), Ocean Tide, AuthorHouse Publishing.
- Millero F J (2005), Chemical Oceanography, Third Edition, CRC Press.
- Bascon W (2002) Waves and Beaches: The Dynamics of the Ocean Surface, Doubleday, New York.
- Morelock (2002), Geological Oceanography, Macmillan Library Reference.
- Jon Erickson & Timothy M. Kusky, (2003), Marine Geology: Exploring the New Frontiers of the Ocean, Checkmark Books
- Kennet J P (1999), Marine Geology, Englewood Cliffs, Prentice – Hall, New Jersey.

COURSE OUTLINES FOR FOURTH SEMESTER

SPA 602: Solar and Ionospheric Physics (3+0)

Magneto-ionic theory, Maxwell's equations, propagation of electromagnetic waves in isotropic medium, constitutive relations for anisotropic medium, polarization, phase and group velocities, solar atmosphere, structure of sun, motion of charged particles in magnetic field, solar oscillations, convection and rotation, solar wind and heliosphere, solar eruptions.

Ionosphere and radio wave propagation, plasma and Alfvén waves, formation of Chapman layers, ion chemistry, Appleton-Hartree equation and its applications in ionosphere, steady-state conductivity of ionosphere, ionospheric phenomena and measurements, auroras, conversion of vertical to oblique incidence, ionogram scaling techniques, use of incoherent data for ionospheric research, HAARP.

Recommended Books:

- Fineschi S and Viereck R A (2007), Solar Physics and Space Weather Instrumentation, Society of Photo Optical.
- Zaprowski B (2005), Earth and Beyond: An Introduction to Earth-Space Science Laboratory Manual, Third Edition, Kendall/Hunt Publishing Company.
- Severny A (2004), Solar Physics, University Press of the Pacific.
- Schunk R W and Nagy A F (2004), Ionospheres: Physics, Plasma Physics, and Chemistry, Cambridge University Press.
- Hargreaves J.K. (1995), The Solar Terrestrial Environment, Cambridge University Press, U.K.
- Colloquium C and Kuo F (1994), Low-Latitude Ionospheric Physics, Pergamon Press Inc.
- Davies K (1990), Ionospheric Radio, The Institution of Engineering and Technology.
- Schunk R and Nagy A (2009), Ionospheres: Physics, Plasma Physics, and Chemistry, Second Edition, Cambridge University Press.
- Kelley M C (2009), The Earth's Ionosphere, Volume 96, Second Edition: Plasma Physics & Electrodynamics, Second Edition, Academic Press.

SPA 604: Astrophysics (3+0)

Content of solar system, Formation of solar system, Roche lobe, Exo-planets, Radiative transfer, Stellar Atmosphere, Stellar opacity, Hydrostatic equilibrium, Fundamentals Equations of Stellar structure, structure of Sun, Photosphere, Chromospheres, Corona, Solar Wind, Magnetic Field of the Sun, Solar Cycle, Sunspot, Limb darkening, solar Activity, Distances to Star, HR diagram, Milky way galaxy, Interstellar Dust and Gas, Interstellar Chemistry, Virial theorem in stars formation, Accretion disk of protostars, Hayashi tracks in stellar evolution, Zero Age Main Sequence Stars(ZAMS), HI and HII regions, open and globular clusters, Pulsating Stars, Stellar Motion, Evolution of Stars, Star Death, Chandrasekhar Limit, Types of Supernovae, white dwarf, Neutron stars, Pulsars, Black Holes, Evolution of Our Galaxy, Classification of Galaxies, Cluster of Galaxies, Dark Matter, Dark Energy, Big Bang Theory, accelerating Universe, fate of the Universe.

Recommended Books:

- Ryden B and Peterson B M (2009), Foundations of Astrophysics, Addison Wesley.
- Irwin J A (2007), Astrophysics: Decoding the Cosmos, Wiley.
- Choudhuri A R (2010), Astrophysics for Physicists, Cambridge University Press.
- Zeilik M (2002), Astronomy: The Evolving Universe, Ninth Edition, Cambridge University Press.
- Harwit M (2006), Astrophysical Concepts, Fourth Edition, Springer.
- Mason J W (2006), Astrophysics Update 2, Springer.

- Carroll B W and Ostlie D A (2006), An Introduction to Modern Astrophysics, Second Edition, Benjamin Cummings.
- Maoz D (2007), Astrophysics in a Nutshell, Princeton University Press.
- Gregory S A and Zeilik M (1997), Introductory Astronomy and Astrophysics, Fourth Edition, Brooks Cole.

SPA 606: Gravitation Physics (3+0)

Transformation Law for Velocity and Acceleration, Minkowski Space, Four-Vectors, Intervals and Light Cone, Proper Time, Position 4-Vector, 4-Velocity and 4-Momentum, 4-Force and Equation of Motion, Motion under a Constant Force, Time Dilation, Length Contraction, Mass Variation, Einstein's Mass-Energy Relationship and its Practical Examples, Motion of a Charged Particle in Transverse Electric and Magnetic Fields, Particles of Zero Rest Mass, Compton Effect, Pair Production, Matter Annihilation, Emission of a Photon from an Excited Nucleus, Variable Proper Mass, Relativistic Optics, Doppler Effect, Aberration of Light, Transformations of Electromagnetic field, Non-inertial Frames Consequences, Tachyons.

Manifolds and Coordinates, Parallel Transport, Tensors, Covariant Derivatives of a Vector and Tensor, Geodesic Equations, Affine Parameters, Bianchi Identity, Equivalence Principle, Riemann Tensor, Ricci Tensor, Energy-Momentum Tensor, Einstein Field Equations, Schwarzschild Solution of Einstein Field Equations, Advance of Perihelion, Deflection of light ray, Gravitational Waves.

Recommended Books:

- Cheng T P (2010), Relativity, Gravitation and Cosmology: A Basic Introduction, Second Edition, Oxford University Press, USA.
- Schutz B (2009), A First Course in General Relativity, Second Edition, Cambridge University Press.
- Taylor and Wheeler, (1992), Spacetime Physics: introduction to special relativity, Second Edition, W. H. Freeman, New York.
- Woodhouse (1992), Special relativity, Springer-Verlag, Berlin.
- Helliwell T M (2009), Special Relativity, University Science Books.
- Rindler W (1991), Introduction to Special Relativity, Second Edition, Oxford University Press, USA.
- Rindler W (2006), Relativity: Special, General, and Cosmological, Second Edition, Oxford University Press, USA.
- Stephani H., (2004), General Relativity & Introduction to Gravitation Fields, Third Edition, Cambridge University Press.

Course Outlines for Optional Subject-III & IV

SPA 608: Space Mission Design (3+0)

Fundamentals of systems engineering, identification and problems definition, Synthesis, analysis, and evaluation activities during conceptual and preliminary system design phases, Articulation through examples and case studies.

Real-world application of the entire space systems engineering discipline, Basic mission objectives and principles and practical methods for mission design and operations in depth. Interactive discussions focus on initial requirements definition, operations concept development, architecture tradeoffs, payload design, bus sizing, subsystem definition, system manufacturing, verification and operations. This is a hands-on course with a focus on robotic missions for science, military and commercial applications.

Recommended Books:

- Larson W J and Wertz J R (1999), Space Mission Analysis and Design, Third Edition, Microcosm.
- Roddy D (2006), Satellite Communications, Fourth Edition, McGraw-Hill Professional.
- Larson W J and Pranke L K, (1999), Human Spaceflight: Mission Analysis and Design, McGraw-Hill Companies.
- Wertz J R, (2001), Mission Geometry: Orbit and Constellation Design and Management - Spacecraft Orbit and Attitude Systems, Springer.
- Brown C D et al. (1998), Spacecraft Mission Design, Second Edition, AIAA (American Institute of Aeronautics & Ast.).
- Kemble S (2006), Interplanetary Mission Analysis and Design, Springer.
- Wertz J R & Larson W J (1996), Reducing Space Mission Cost, Springer.
- Wertz J R, Everett D F and Puschell J J (2011), Space Mission Engineering: The New SMAD, Microcosm Press.
- Griffin M D and French J R (2004), Space Vehicle Design, Second Edition, AIAA (American Institute of Aeronautics & Ast.).
- Pocha J J (1987), An Introduction to Mission Design for Geostationary Satellites, Springer.

SPA 610: Air & Space-Borne Sensors (3+0)

Theoretical and practical issues associated with applying modern, airborne remote sensing technologies to precision mapping problems. Navigation, LANDSAT, SAR, and hyperspectral data.

Recommended Books:

- Abid M M (2005), Spacecraft Sensors, Wiley.
- Nebylov A (2011), Aerospace Sensors, Momentum Press.
- Jensen J R (2005), Introductory Digital Image Processing: A Remote Sensing Perspective, Third Edition, Prentice Hall, Toronto, Canada.
- Jensen J R (2007), Remote Sensing of Environment and Earth Resources Perspective, Second Edition, Prentice Hall, Toronto, Canada.
- Elwenspoek M (2001), Sensor Technology 2001, Springer.
- Imbriale W A and Yuen J H (2006), Spaceborne Antennas for Planetary Exploration, Wiley-Interscience.
- Merhav S (1998), Aerospace Sensor Systems and Applications, Springer.
- Cutts J A (1994), Advance micro-devices & Space Science Sensors, Society of Photo Optical.

SPA 612: Aerospace and their Applications (3+0)

Aerospace Systems Overview, System Elements and Components, System Program and Operational Organization, Space Segment, Post-Launch Requirements, Satellite Composition, Critical Structural Phases, Mission and Satellite Control Activities, Satellite Orbits: Foundation, Equation of Satellite Orbit, Satellite Path in Space, The Two-Body Problem, Classical Orbital Elements, The Geostationary Orbit, Change of Longitudes, Orbital Perturbations, Solar Radiation Pressure, Atmospheric Drag, Types of Orbits, Geometry of Geostationary Orbit, Basics of Communications, Launch of Geostationary Satellite (ELV, Space Shuttle) and concept of Launch Window, The Non-Spherical Earth, Eclipse Geometry, Sun Interference. Launch Vehicles, Rocket Propulsion, Chemical Propulsion, Solid Propellants, Multi Stage Rocket, Injection into Final Orbit, Launch Vehicles for Commercial Satellites, Attitude Control, Propulsion Subsystem, System Reliability, Estimation of Mass, Telemetry Interface Functions, Telecommand Interface Functions, Tracking Interface Functions, Satellite Applications: Communications (Civil and Defence), Navigation (Transit, Navstar GPS), Surveillance (Civil and Military), Meteorology, DMSP, METEOR, FENGYUN, METEOSAT AND GMS. Earth Resources /Imaging Satellite, BADAR I & II, PAKSAT I & II, TERRA, METEOR, and Advanced Satellites, Recent Developments in Space Systems. Search and Rescue: COSPAS-SARSAT, National Missile Defence Program (NMD), Anti-Satellite Weapon System etc.

Recommended Books:

- Pratt T, Bostian C W and Allnutt J E (2002), Satellite Communication, Second Edition, John Wiley and Sons, New York.
- Garner J T and Jones M (1990), Satellite Operations: Systems Approach to Design and Control, Ellis Horwood Ltd.
- Pritchard W, Suyderhoud H and Nelson R A (1993), Satellite Communication System Engineering, Second Edition, Prentice Hall.
- Davidoff M (1994), The Satellite Experimenter's Hand Book, The American Radio Relay League.
- Richharia M (1999), Satellite Communication Systems, Second Edition, McGraw-Hill Professional.
- Garis D D, Winterton R, Harding R and Dutton L (1990), Military Space (Brassey's Air Power: Aircraft, Weapons Systems and Technology), Brasseys Publications.
- Fthenakis E (1984), Manual of Satellite Communications, McGraw-Hill Co., USA.
- Wertz J R (1980), Spacecraft Attitude Determination and Control, D. Reidel Publishing Company.

SPA 614: Environmental Science (2+1)

Introduction: basic concepts, history, nature and scope of Environmental Science and its contribution to society. Different aspects of environment: physical, ecological, socio-economic, ethical, philosophical. Major components of environment: physico-chemical, biological and social, and their relationships with various environmental factors. Human environment and its problems: global, national, regional. Environmental challenges for sustainable development: current and future trends in population growth, environmental pollution, development in industry and agriculture, urbanization, poverty and resource depletion, globalization and environmental issues, Science and the globalization of environmental

discourse, Aerosol, Urban Smog, Green house gases, Global warming, Ozone and Climate change, Air Trajectories.

Labs:

Study of the spectrum of light. Structure and function of spectrophotometer for absorption and transmission of light. Measurement and comparison of the light intensity at different locations and times using lux meter. Measurement the noise level at different location to assess the noise pollution using sound level meter. Collection of particulate matter and its measurements by gravimetric methods. Sampling techniques (air, water and soil) for physical and chemical monitoring, Study the indicators for biological monitoring of the river and canal water, the indicators for ecological monitoring in the field for fauna and flora, use of various instrumental techniques for analysis of samples.

Recommended Books:

- Cunningham W and Cunningham M (2010), Principles of Environmental Science: Inquiry & Applications, Sixth Edition, McGraw-Hill Science/Engineering/Math.
- Botkin D B and Keller E A (2007), Environmental Science: Earth as a Living Planet, Sixth Edition, John Wiley & Sons.
- Miller G T and Spoolman S (2012), Environmental Science, Fourteen Edition, Brooks Cole.
- McKinney M L, Schoch R M and Yonavjak, L (2007), Environmental Science: Systems and Solutions, Fourth Edition, Jones & Bartlett Publishers.
- Wright R T and Nebel B J (2007), Environmental Science: Toward a Sustainable Future, Tenth Edition, Pearson Educational.
- Miller G (2002), Environmental Science: Working with the Earth, Thomson Learning.
- Hill M K (2004), Understanding Environmental Pollution, Second Edition, Cambridge University Press, UK.
- Botkin D B and Keller E A (2007), Environmental Science: Earth as a Living Planet, Sixth Edition, Wiley.
- Cunningham W and Cunningham M (2009), Environmental Science: A Global Concern, Eleventh Edition, McGraw-Hill Science/Engineering/Math.
- Wright R T and Boorse D (2010), Environmental Science: Toward a Sustainable Future, Eleventh Edition, Addison Wesley.
- Friedland A, Relyea R and Courard-Hauri D (2011), Environmental Science: Foundations and Applications, H. Freeman.
- Monteith J and Unsworth M (2007), Principles of Environmental Physics, Third Edition, Academic Press.
- Hughes P (2001), Introduction to Environmental Physics, Taylor & Francis.
- Smith C (2001), Environmental Physics, Routledge.
- Loxsom (2000), Environmental Physics, John Wiley & Sons Inc.
- Boeker E and Grondelle R V (2000), Environmental Physics, Second Edition, Wiley.
- Alibrandi M (2003), GIS in the Classroom: Using Geographic Information Systems in Social Studies and Environmental Science, Heinemann.
- Aspinall R (2003), Environmental GIS and Data Analysis, John Wiley & Sons Inc.
- Lovett, (2001), GIS in the Environmental Sciences, Blackwell Science, UK.
- Douglas W (1994), Environmental GIS Applications to Industrial Facilities, CRC Press.

SPA 616: Renewable Energy Resources (3+0)

Overview of renewable energy resources, the basic principles of solar electricity, solar water heating, small-scale wind power, micro hydro-power, bioenergy, heat pumps and their applications for homes,

businesses and farms, Study of the main renewable energy technologies, biomass, geothermal, hydroelectric, solar thermal, photovoltaic, marine energy resources and wind, including technical and economical issues, available energy resources and their associated environmental and climatological impacts and the potential and limitations of renewable energy technologies.

Recommended Books:

- Fornasiero P and Mauro Graziani, (2011), Renewable Resources and Renewable Energy: A Global Challenge, Second Edition, CRC Press.
- Tiwari G N and Ghosal M K (2005), Renewable Energy Resources: Basic Principles and Applications, Alpha Science International, Ltd.
- Twidell J and Weir T (2005), Renewable Energy Resources, Second Edition, Taylor & Francis.
- Singh M P (2004), Natural Resources and Renewable Energy, Daya Publishing House.
- Jeffrey S B (2011), The Renewable Energy Debate: The Commercialization of Renewable Resources, Webster's Digital Services.
- Boyle G (2004), Renewable Energy: Power for a Sustainable Future, Second Edition, Oxford University Press.
- da Rosa A V (2009), Fundamentals of Renewable Energy Processes, Second Edition, Academic Press.

SPA 618: Costal Zone Management (3+0)

Introduction to Oceanography, Origin of Earth and Oceans, Plate Tectonics, The Sea Floor, Marine Sediments, Ocean Water Chemistry, Physical Properties of Water, Physics of the Ocean, The Ocean in Motion, Waves, Tides, Beaches and Coastal Geology, Estuaries, Life in the Ocean, Productivity, The coast and coastal issues, The natural system and physical processes, Boundaries of the coastal zone, problems and challenges in the Coastal Zone, areas of environmental concern, shoreland systems, Estuarine systems, Nearshore and ocean systems, Introduction to coastal & sea resources, their uses, Conserving and Restoring Coastal Habitats, Wetlands along Pakistan coast, Renewable and non-renewable resources, use of resources, Coastal Ecosystem and Resources, Coastal Zone Management Techniques, Water Quality Management in the Coastal Zone, Management of Beaches & Coastal Hazards, Marine and Coastal Protected Area Management. Institutional and Legal Arrangements, Remote sensing application for coastal erosion and accretion, salinity intrusion, navigation, morphology of the sea, biological productivity of ocean environment, Use of satellite data to performing wetland mapping, site location for mariculture. Sea surface temperature studies to establish links with the physical, biological and chemical processes of ocean and atmosphere, Mangroves and coral reefs, fish population, and pollution assessment-case studies.

Recommended Books

- Beatly T, Brower, D, and Schwab A. K. (2002), An introduction to Coastal Zone Management, island Press.
- Vernberg, W. B. (2001). The coastal zone: Past, present and future, Univ. of South Carolina Press.
- Gower J (2007), Remote Sensing Oceanography, third edition, John Wiley & sons,
- Bartlett D. and Smith J (2004). GIS for Coastal Zone Management, CRC Presss.

- Robinson I S. (2004). Measuring the ocean from space: The Principals and methods of satellite oceanography, Springer.
- Timothy, B., Brower, D. J. and Schawab, A. K. (2002). An Introduction of coastal zone management, 2nd edition, Island Press.
- Carter, B. (1988). Coastal Environments, Academic Press.
- Clark, J. R. (1996). Coastal Zone Management Handbook, CRC Press.
- Clark, T. (1977). Coastal Ecosystem Management, John Wiley & Sons,
- Salm, R. V. and Clark, T (1984). Marine and Coastal Protected Area: A guide for planners and managers, IUCN.
- Snadeker, P. and Getter, E. (1995). Coastal Resources Management Guidelines Coastal Man. Publ. no 2, National Park Service, U.S. Department of Interior.
- Snedaker, S.C. and Getter, C.D. (1985). Coasts: Coastal Resources Management Guidelines. Coastal Publication No.2 205pp.
- Sorensen, J. C., McCreary S.T. and Hershman M.J. (1984). Institutional Arrangements for Management of Coastal Resources. Coastal Publication No.1 165pp.
- UNITED NATIONS. (1995). Guidelines on Environmentally Sound Development of Coastal Tourism. New York. 124 pp.
- Hord R. M. (1986), Remote sensing methods and application, John Wiley and sons, New York, USA.

SPA 620: Natural Hazards & Disaster Management (3+0)

Introduction to the earth's structure, and the processes that occur inside the earth and at the earth's surface, several types of natural hazards, case studies of populations effected by natural disasters, the possible ways in which hazards may be mitigated, and the response of local public officials, members of society and the global community to catastrophic events. Topics covered in the course will include: structure of the earth, energy sources that affect processes that occur on the planet, plate tectonics, hazard versus risk, global distribution of geologic hazards and populations at risk, volcanoes, earthquakes, tsunamis, floods, Hurricanes, El Niño, tornadoes, subsidence, landslides, ice storms, avalanches, severe weather events, meteor impacts.

Disaster management, disaster mitigation, past responses to natural disasters, prediction of natural disasters and planning for future disasters.

Education on disasters, Community involvement, The adjustment of Human Population to Natural hazards & disasters, Role of Media: Monitoring Management- Discuss the programme of disaster research & mitigation of disaster of following organizations: International Council for Scientific Unions (ICSU)- Scientific committee on problems of the Environment (SCOPE), International Geosphere-Biosphere programme (IGBP), World federation of Engineering Organizations(WFED), World Meteorological organizations(WMO), Geographical Information System(GIS), International Association of Seismology & Physics of Earth's Interior (IASPEI), Various U.N agencies like UNCRD, IDNDR, WHO, UNESCO, UNICEF, UNEP, Environmental policies & programmes in Pakistan- Environmental Legislations in Pakistan, Awareness, Conservation Movement, Education & training.

Recommended Books:

- Abbott P L (2011), Natural Disasters, Eighth Edition, McGraw Hill Publishing.
- Abbott, P. L., (2001), Natural Disasters, McGraw-Hill, Third Edition.
- Bluestein H. B., (1999), Tornado Alley: Monster Storms of the Great Plains, Oxford University Press.
- Fisher R.V., G. Heiken, and J.B. Hulen, (1997), Volcanoes: Crucibles of Change, Princeton, Princeton University Press.
- Frater H (1999), Natural disasters, Berlin, Springer-Verlag.

- French B M (1998), *Traces of Catastrophe*, Houston, Lunar and Planetary Institute.
- Ingleton, J, ed., (1999), *Natural Disaster Management*, Paris, UNESCO.
- Larson E (2000), *Isaac's Storm: A Man, a Time and the Deadliest Hurricane in History*, Vintage.
- McCall G.J.H., D.J.C. Laming and S.C. Scott, eds., (1992), *Geohazards: Natural and Man-Made*, London, Chapman and Hall.
- McGuire B., I. Mason, and C. Kilburn, (2002), *Natural Hazards and Environmental Change*, London, Arnold.
- Rauber R., J. Walsh, and D. Charlevoix (2002), *Severe and Hazardous Weather*, Kendall/Hunt.
- Sieh K., and S. LeVay, (1998), *The Earth in Turmoil: Earthquakes, Volcanoes and Their Impact on Humankind*, New York, W.H. Freeman.
- Simon S., (1999), *Tornadoes*, HCP.
- Toomey D (2002), *Storm Chasers: The Hurricane Hunters and Their Fateful Flight into Hurricane Janet*. W. W. Norton and Co.
- Zebrowski E (2002), *Perils of a Restless Planet: Scientific Perspectives on Natural Disasters*, Cambridge, Cambridge University Press.
- Courtillot, V (1999), *Evolutionary Catastrophes*, Cambridge, Cambridge University Press.
- Keller E A and DeVecchio D E (2011), *Natural Hazards; Earth's Processes as Hazards, Disasters, and Catastrophes*, Third Edition, Prentice Hall.
- Bryant E (2005), *Natural Hazards*, Cambridge, Cambridge University Press.
- Chester D (1995), *Volcanoes and Society*, Halsted Press.
- Kanti Paul B K (2012), *Environmental Hazards and Disasters: Contexts, Perspectives and Management*, Wiley.
- Singh R B (2006), *Natural Hazards and Disaster Management, Vulnerability and Mitigation*, Rawat.
- Jat B C (2008), *Natural Hazards & Disaster Management*, MD Pub Pvt. Ltd.

SPA 628: Project (0+3)

SPA 630: Thesis (0+6)